Code and Analysis Repository

The LMS item asks for a link or a zip of notebooks. This project submits the link. The analysis is Python, tested with pytest.

https://github.com/aushakanuj/education-platform

1. Link

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Owner on GitHub: aushakanuj. Default branch: main. This Submission Package is added on a branch and opened as a pull request so the files can be reviewed; the link above is the project the report describes.

2. What “analysis” means here

There is no notebook bundle. The analysis is a small rule engine with a deterministic generated school behind it, plus tests that lock the worked cases in the report.

Role	Path
Generated school	backend/src/education_platform/modules/synthetic/
Rebuild command	From backend/: uv run python -m education_platform.modules.synthetic.cli
Seed and size	SchoolSpec in generator.py. Seed 20260812. 240 students by default.
Early-warning rules	backend/src/education_platform/modules/at_risk/engine.py
Loading, storage, permissions	backend/src/education_platform/modules/at_risk/service.py and router.py
student_360 register	backend/src/education_platform/modules/insights/
Policy the rules implement	docs/design/08-at-risk-early-warning.md
Engine tests	backend/tests/test_at_risk_engine.py
API, role, and audit tests	backend/tests/test_at_risk_api.py
Teacher list matches the assistant	backend/tests/test_at_risk_assistant_agreement.py
Schema the questions are written against	backend/src/education_platform/modules/text_to_sql/config/schema_catalog.md
How to run the whole app	README.md at the repository root

3. Run the analysis

1. Install uv, Docker, and Node 18 or newer.
2. From the repository root: docker compose up -d postgres. Wait until it is healthy.
3. From backend/: uv run alembic upgrade head
4. From backend/: uv run python -m education_platform.modules.synthetic.cli
5. From backend/: uv run pytest tests/test_at_risk_engine.py tests/test_at_risk_api.py tests/test_at_risk_assistant_agreement.py
6. Optional, the app itself: uv run uvicorn education_platform.main:app --reload --host 127.0.0.1 --port 8000, and from frontend/ npm run dev after copying .env.example to .env.

The engine tests include cases that do not need a database. The API and agreement tests need the migrated Postgres from step 2. CI on GitHub runs the full backend suite against a pgvector container.

4. What not to upload beside the link

- backend/.env if it contains an API key.
- The local upload directory of PDF blobs.
- A dump of a database, even the synthetic one. The seed rebuilds it.
- node_modules or the Python virtual environment.

A zip of the whole working tree would be a worse copy than the link: it goes stale, and it is easy to sweep a secret into it. If the LMS form will not accept a link without a file, upload this note (PDF and DOCX) and paste the link in the form field.